ELECTRIC VEHICLE TECHNOLOGY INTERNSHIP

TASK 4

EV Design & Energy Project

Conceptual design of an electric commuter scooter

A calculation-based preliminary design covering vehicle requirements, motor selection, battery capacity, estimated driving range, and energy consumption under different operating conditions.

Prepared by	**Ali Ashab Shah**
Academic profile	3rd Year Mechanical Engineering Student
Internship platform	SkillZeno
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Design note: This report is an educational, first-pass design. Values are engineering estimates based on stated assumptions, not certified vehicle specifications. A production design would require simulation, component datasheets, prototype testing, and applicable safety homologation.

Project objective: develop a practical and transparent baseline design that can be checked, improved, and validated with real component data.

1. Executive Summary

This project proposes a battery-electric commuter scooter for daily urban travel. The design target is a practical city vehicle with a nominal top speed near 60 km/h, a useful real-world range around 70 km in mixed riding, and a battery that can supply short acceleration peaks without being sized only for peak power.

Design item	Preliminary selection / target
Vehicle application	Single-rider urban commuter scooter
Assumed loaded mass	180 kg (vehicle + rider + small cargo)
Maximum design speed	60 km/h
Motor	3.0 kW continuous BLDC/PMSM; approximately 5 kW peak
Battery	72 V nominal, 40 Ah, 2.88 kWh nominal energy
Usable battery energy	Approximately 2.45 kWh at 85% usable fraction
Estimated mixed-use range	Approximately 70 km, depending on conditions
Charging concept	External isolated charger matched to pack voltage and BMS

2. Design Requirements and Assumptions

The following requirements are selected for a representative commuter scooter. They are design inputs for calculations, not measured performance claims.

Parameter	Assumption	Reason
Loaded mass, m	180 kg	Vehicle, rider and light cargo
Target top speed	60 km/h = 16.67 m/s	Urban/suburban mobility target
Wheel radius, r	0.25 m	Representative scooter wheel radius
Rolling resistance coefficient, Crr	0.015	Approximate road-tire value
Aerodynamic drag coefficient, Cd	0.9	Upright rider and scooter estimate
Frontal area, A	0.60 m^2	Approximate projected area
Air density, ρ	1.225 kg/m^3	Near sea-level reference
Drivetrain efficiency	90%	Motor, inverter and mechanical losses combined
Auxiliary load	80 W average	Lights, display, control electronics, etc.

3. Motor Requirement and Powertrain Selection

The motor must overcome rolling resistance and aerodynamic drag at the target speed. Acceleration and gradients require additional power, so the continuous rating is not selected from top-speed cruising alone.

3.1 Road-load calculation at 60 km/h

Rolling resistance force

$F_{rr} = Crr \times m \times g = 0.015 \times 180 \times 9.81 =$ **26.5 N**

Aerodynamic drag force

$F_d = \frac{1}{2} \times \rho \times Cd \times A \times v^2$

$= 0.5 \times 1.225 \times 0.9 \times 0.60 \times (16.67)^2 =$ **45.9 N**

Total steady road-load force

$F_{total} = F_{rr} + F_d = 26.5 + 45.9 =$ **72.4 N**

Wheel power

$P_{wheel} = F_{total} \times v = 72.4 \times 16.67 =$ **1.21 kW**

Electrical power for traction

$P_{elec,traction} = P_{wheel} / \eta = 1.21 / 0.90 =$ **1.35 kW**

Adding approximately 80 W for auxiliary loads gives about 1.43 kW at steady 60 km/h under the stated flat-road assumptions. Real power will be higher with headwinds, hills, heavier loads, low tire pressure, or frequent acceleration.

3.2 Acceleration and grade allowance

For a preliminary acceleration check, assume acceleration of 0.8 m/s². The additional inertial force is F = m × a = 180 × 0.8 = 144 N. At 30 km/h (8.33 m/s), inertial wheel power alone is approximately 1.20 kW. This is in addition to road-load power, so a short peak rating is necessary.

A 5 kW peak motor/controller combination provides headroom for acceleration and brief moderate gradients. The exact gradeability and acceleration time must be verified using the selected motor torque curve, controller current limit, wheel diameter, and thermal limits.

Subsystem	Proposed specification	Selection justification
Traction motor	72 V BLDC/PMSM; 3.0 kW continuous, ~5 kW peak	Continuous output exceeds flat-road cruise demand; peak margin supports launch and transient loads.
Motor controller	72 V compatible; current-limited; regenerative-braking capable if supported	Must match motor phase current, battery voltage and thermal rating.
Final drive	Hub motor or reduction drive, selected with motor supplier	Ratio/torque must be matched to wheel size and target speed.
Braking	Independent mechanical/hydraulic service brakes	Regeneration is supplementary and must not replace service brakes.

4. Battery Capacity and Estimated Range

4.1 Battery sizing

Selected pack: 72 V nominal × 40 Ah = **2,880 Wh = 2.88 kWh nominal energy**.

To avoid assuming the entire battery can be used on every trip, an 85% usable-energy factor is applied:

$$E_{usable} = 2.88\ kWh \times 0.85 = 2.448\ kWh \approx 2.45\ kWh$$

The 85% factor is a planning assumption for operating reserve and battery-life margin. The battery management system (BMS) must enforce cell voltage, current, temperature and balancing limits. The pack chemistry, cell configuration and exact voltage window must be chosen from supplier datasheets.

4.2 Range calculation

Range is estimated from usable battery energy divided by energy consumption per kilometre. For a mixed-use baseline, assume 35 Wh/km at the battery, including typical stop-start riding and auxiliaries.

Range = 2,448 Wh ÷ 35 Wh/km = 69.9 km. Therefore, the indicative mixed-use range target is about 70 km.

This is a calculated estimate, not a guaranteed range. Actual range depends on speed, rider mass, ambient temperature, tire pressure, road slope, wind, battery age, traffic, and riding style.

4.3 Battery current check

At 3.0 kW electrical input and 72 V nominal voltage, the approximate current is I = P/V = 3,000/72 = **41.7 A**. At 5.0 kW peak, I ≈ 5,000/72 = **69.4 A**. Because pack voltage varies and losses/current peaks occur, the BMS, cells, busbars, connectors and fuse must be rated from the actual maximum-current specification, with suitable safety margin.

Battery item	Preliminary value	Design note
Nominal pack voltage	72 V	Actual voltage window depends on chemistry and series count.
Capacity	40 Ah	Confirm continuous and peak discharge ratings with cell supplier.
Nominal energy	2.88 kWh	72 V × 40 Ah
Planning usable energy	2.45 kWh	Assumes 85% usable fraction
Estimated mixed range	~70 km	At 35 Wh/km battery-side consumption

5. Energy Consumption Under Different Conditions

Energy use changes substantially with speed and riding pattern. The following are illustrative planning scenarios for the same 2.45 kWh usable battery. They are estimates for comparison and should be validated by road tests.

Usage condition	Assumed consumption	Estimated range	Reason
Gentle city riding	28 Wh/km	2,448/28 = 87 km	Moderate speeds and smooth acceleration
Mixed urban commute (baseline)	35 Wh/km	2,448/35 = 70 km	Stops, starts, cruising and auxiliaries
Fast riding / higher average speed	45 Wh/km	2,448/45 = 54 km	Aerodynamic drag increases strongly with speed
Hilly route / heavy stop-start	50 Wh/km	2,448/50 = 49 km	Repeated acceleration and climbing increase demand
Cold weather / aged battery planning case	55 Wh/km equivalent	2,448/55 = 45 km	Higher losses and reduced available energy may occur

Key interpretation

The table shows why range should be reported as a band rather than one fixed number. At higher speeds, aerodynamic drag rises approximately with the square of speed, and the power needed to overcome it rises approximately with the cube of speed. Repeated acceleration also raises energy use, although regenerative braking may recover some energy where supported and conditions permit.

5.1 Simple energy budget for a 50 km mixed commute

At the baseline consumption of 35 Wh/km, a 50 km trip uses approximately 50 × 35 = **1,750 Wh (1.75 kWh)**. Compared with 2.45 kWh of planning-usable energy, the trip would use about 1.75/2.448 × 100 = **71.5%** of usable energy, leaving roughly 0.70 kWh as reserve under the assumptions.

5.2 Charging concept

A compatible off-board charger should be selected for the exact battery chemistry and maximum pack voltage. For a rough charging-time estimate, a 500 W charger delivering an average of 0.45 kW to the battery would require 2.88 kWh / 0.45 kW ≈ 6.4 hours from near-empty to nominal full energy. Allow additional time for charge taper, balancing, temperature limits and conversion losses. Charging must follow the battery and charger manufacturers' instructions.

6. System Architecture, Safety and Validation

6.1 Basic system architecture

Energy / signal path	Function	Important checks
Battery pack → main fuse → contactor / disconnect	Stores and safely isolates high-voltage energy	Fuse coordination, insulation, service disconnect and enclosure protection
Battery pack ↔ BMS	Monitors cell voltage and temperature; protects pack	Over/under-voltage, overcurrent, overtemperature and cell balancing
Battery → controller → motor	Converts DC energy into controlled motor torque	Voltage/current compatibility, thermal limits, waterproofing and secure wiring
Throttle, brake switches and vehicle controls	Provides rider commands and interlocks	Fail-safe response, unintended-throttle prevention and brake cut-off logic
Charger → battery charging port	Restores battery energy	Correct chemistry/voltage profile, connector protection and charge interlock

6.2 Safety considerations

• Use a battery pack designed and tested for traction applications; do not assemble or modify a high-energy pack without qualified supervision.
• Include a correctly rated fuse, contactor or disconnect, BMS, protected connectors, insulation, strain relief and a mechanically robust enclosure.
• Keep the battery protected from water ingress, impact, vibration and excessive heat; provide safe service access.
• Match charger, BMS, cells, controller and motor to the same voltage/current limits.
• Validate braking, electrical isolation, thermal behaviour, electromagnetic compatibility and applicable local vehicle regulations before any road use.

6.3 Validation plan

Test	Measurement	Acceptance approach
Bench motor test	Voltage, current, speed, torque and temperature	Verify continuous and peak ratings remain within supplier limits
Energy consumption route	Wh/km over a repeatable route	Record rider/load, speed profile, weather and tire pressure
Range test	Distance to defined low state-of-charge threshold	Repeat several runs and report average plus variation
Thermal test	Battery, controller and motor temperatures	Remain below manufacturer limits in worst expected duty
Charging test	Charge time, energy from wall and pack temperature	Confirm charger/BMS compatibility and safe cut-off behaviour

7. Conclusion

A preliminary electric commuter scooter design has been developed using transparent assumptions and first-order calculations. A 72 V, 40 Ah battery provides 2.88 kWh nominal energy, with approximately 2.45 kWh used for planning. Road-load calculations estimate about 1.43 kW electrical demand at steady 60 km/h on level ground under the chosen assumptions. A 3.0 kW continuous motor with approximately 5 kW peak capability is a reasonable starting point for component selection, subject to supplier data and validation. Estimated range is about 70 km for mixed riding, with scenario estimates from roughly 45 to 87 km depending on conditions.

The next engineering step is to select real motor, controller, battery-cell and charger datasheets, then validate performance through simulation and controlled tests. This will convert the conceptual design into a defensible,

application-ready specification.

References and calculation basis

1. Fundamental vehicle road-load equations: rolling resistance Frr = Crr·m·g; aerodynamic drag Fd = ½·ρ·Cd·A·v²; wheel power P = F·v.
2. Electrical energy relation: E = V·Ah; estimated range = usable energy / energy consumption per distance.
3. All numeric vehicle parameters in this report are stated design assumptions for this educational project. They should be replaced by validated component and test data during detailed design.